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<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8">
<title>TradeMatrix – Data Schema</title>
<style>
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  body { font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial, sans-serif; color: #2D3748; line-height: 1.4; font-size: 9pt; }
  h1 { color: #1A365D; font-size: 18pt; border-bottom: 2px solid #2B6CB0; padding-bottom: 6px; }
  h2 { color: #2B6CB0; font-size: 12pt; border-bottom: 1px solid #E2E8F0; padding-bottom: 4px; margin-top: 15px; }
  table { width: 100%; border-collapse: collapse; margin-top: 8px; font-size: 8.5pt; }
  th, td { border: 1px solid #CBD5E0; padding: 6px 8px; text-align: left; }
  th { background-color: #EDF2F7; }
</style>
</head>
<body>
  <h1>Data Schema & Feature Pipeline Specification</h1>
  <h2>1. Ingested Market Schema (raw_market_data)</h2>
  <table>
    <tr><th>Field</th><th>Type</th><th>Description</th><th>Validation Rule</th></tr>
    <tr><td>Date</td><td>Timestamp</td><td>Trading session index (UTC)</td><td>Monotonic increasing, no duplicates</td></tr>
    <tr><td>Open, High, Low, Close</td><td>Float64</td><td>Adjusted OHLC prices</td><td>High &ge; max(Open, Close), Low &le; min(Open, Close)</td></tr>
    <tr><td>Volume</td><td>Int64/Float64</td><td>Total session volume</td><td>Volume &ge; 0</td></tr>
  </table>
  <h2>2. Engineered Features Schema (engineered_features)</h2>
  <table>
    <tr><th>Feature</th><th>Mathematical Formula</th><th>Stationarity</th><th>Leakage Safeguard</th></tr>
    <tr><td><code>log_return</code></td><td>ln(Close_t / Close_{t-1})</td><td>Stationary (p &lt; 0.001)</td><td>Lagged 1 step</td></tr>
    <tr><td><code>target</code></td><td>ln(Close_{t+1} / Close_t)</td><td>Stationary</td><td>Forward shift for training alignment</td></tr>
    <tr><td><code>rsi_14</code></td><td>100 - (100 / (1 + RS_14))</td><td>Bounded [0, 100]</td><td>Trailing exponential window</td></tr>
    <tr><td><code>macd</code></td><td>(EMA_12 - EMA_26) / Close</td><td>Stationary</td><td>Trailing exponential window</td></tr>
    <tr><td><code>volatility_20</code></td><td>Rolling 20-day return std</td><td>Stationary</td><td>Strict trailing 20-day window</td></tr>
    <tr><td><code>volume_zscore</code></td><td>(Vol - Mean20(Vol)) / Std20(Vol)</td><td>Stationary</td><td>Trailing 20-day mean/std only</td></tr>
  </table>
</body>
</html>

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